

Week 6 · Class 1: Practice Exercises — ANSWER KEY

Plotting Data

2025-12-31

1 Non-AI Exercises

1.1 1. Visualization Principles

1.1.1 1.1 Multiple Choice: Chart Types

Which visualization is best for showing the relationship between two continuous variables?

- a) Bar chart
- b) Pie chart
- c) Scatter plot
- d) Box plot

Answer: c) Scatter plot

1.1.2 1.2 Match: Plot Types and Uses

Match each plot type with its best use:

Plot Types:

- a) Histogram
- b) Line chart
- c) Bar chart
- d) Box plot
- e) Heatmap

Uses:

1. Compare categories
2. Show distribution of one variable
3. Display trends over time
4. Show distributions by group
5. Display correlation matrix

Matches: a = 2, b = 3, c = 1, d = 4, e = 5

1.1.3 1.3 True or False: Visualization Best Practices

Mark each statement as True (T) or False (F):

T - Always start y-axis at zero for bar charts **F** - Pie charts are ideal for comparing many categories **T** - Color should be used meaningfully, not decoratively **F** - 3D effects improve data comprehension **T** - Labels should be clear and descriptive

1.2 2. Grammar of Graphics

1.2.1 2.1 Fill in the Blanks: ggplot Components

The main components of a ggplot are:

1. **data** : The dataset
2. **aesthetics (aes)** : Variables mapped to visual properties
3. **geometries (geom)** : The type of plot (points, lines, bars)
4. **labels** : Axis labels and title
5. **theme** : Overall visual style

1.2.2 2.2 What does each line add to the plot?

```
ggplot(data, aes(x = income, y = vote_share)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  facet_wrap(~region) +  
  theme_minimal()
```

Line 2: **Creates scatter plot with points** Line 3: **Adds linear regression line/smooth**
Line 4: **Creates separate panels by region** Line 5: **Applies minimal theme styling**

1.2.3 2.3 Multiple Choice: Aesthetics

In ggplot, aes() can map variables to:

- a) Colors only
- b) X and Y positions only
- c) Multiple visual properties (color, size, shape, etc.)
- d) Statistical transformations

Answer: c) Multiple visual properties (color, size, shape, etc.)

1.3 3. Effective Communication

1.3.1 3.1 Spot the Error: Bad Visualization

What's wrong with using a pie chart to show changes over 50 time periods?

Problem: Pie charts cannot show trends over time; use line chart instead. Too many categories make pie charts unreadable.

1.3.2 3.2 Match: Color Choices

Match color use with purpose:

Color Uses:

- a) Sequential palette
- b) Diverging palette
- c) Categorical palette
- d) Highlight color

Purposes:

1. Show distinct unordered groups
2. Emphasize one category
3. Show values from low to high
4. Show values around a midpoint

Matches: a = 3, b = 4, c = 1, d = 2

1.4 4. Advanced Visualization

1.4.1 4.1 True or False: Faceting

About faceting in ggplot:

T - `facet_wrap()` creates a grid of plots **F** - `facet_grid()` requires two variables (*can use one or two*) **F** - Facets must share the same scale (*can use scales="free"*) **T** - You can combine faceting with grouping **T** - Faceting helps avoid overplotting

1.4.2 4.2 Multiple Choice: Handling Overplotting

When you have too many points overlapping, you should:

- a) Make points smaller
- b) Use transparency (alpha)
- c) Use jittering or binning
- d) All of the above

Answer: d) All of the above

2 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: “Using tidyverse in R, calculate the mean approval_rating by party. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

2.1 5. Election Results Visualization

Dataset: data/election_visualization.csv

Description: State-level election results over time.

Variables:

- state: State name (chr)
- year: Election year (int)
- dem_vote_share: Democratic vote percentage (dbl)
- rep_vote_share: Republican vote percentage (dbl)
- turnout: Voter turnout percentage (dbl)
- swing_state: Swing state indicator (lgl)
- region: Geographic region (chr)
- electoral_votes: Number of electoral votes (int)
- margin: Victory margin (dbl)
- incumbent_party: Party of incumbent (chr)

2.1.1 5.1 Time Series Visualization

```
# Load the dataset
election_data <- read_csv("data/election_visualization.csv")
```

Rows: 250 Columns: 10

-- Column specification -----

Delimiter: ","

chr (3): state, region, incumbent_party

dbl (6): year, electoral_votes, dem_vote_share, rep_vote_share, turnout, margin

lgl (1): swing_state

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Examine the structure
glimpse(election_data)
```

Rows: 250

Columns: 10

\$ state <chr> "Alabama", "Alabama", "Alabama", "Alabama", "Alabama", ~

\$ year <dbl> 2008, 2012, 2016, 2020, 2024, 2008, 2012, 2016, 2020, ~

```

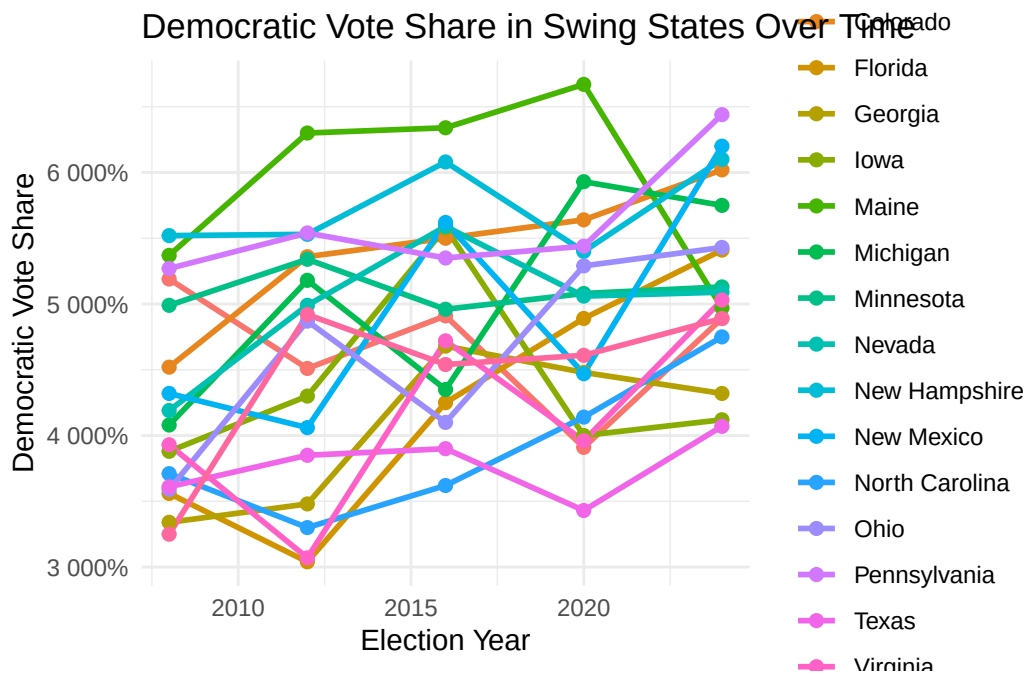
$ region          <chr> "South", "South", "South", "South", "South", "West", "~
$ electoral_votes <dbl> 9, 9, 9, 9, 9, 3, 3, 3, 3, 3, 11, 11, 11, 11, 11, 6, 6~
$ swing_state     <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE~
$ dem_vote_share  <dbl> 28.4, 39.1, 36.0, 34.8, 48.0, 48.3, 49.7, 46.4, 51.8, ~
$ rep_vote_share  <dbl> 70.0, 59.9, 63.8, 64.9, 50.0, 51.5, 49.9, 50.7, 47.6, ~
$ turnout         <dbl> 72.2, 53.5, 57.6, 56.2, 61.3, 65.9, 45.0, 64.0, 68.7, ~
$ margin          <dbl> -41.6, -20.8, -27.8, -30.1, -2.0, -3.2, -0.2, -4.3, 4.~
$ incumbent_party <chr> "Republican", "Democrat", "Democrat", "Republican", "D~

```

```

# Example: Democratic vote share over time in swing states
ggplot(election_data %>% filter(swing_state == TRUE)) +
  aes(x = year, y = dem_vote_share, color = state, group = state) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Democratic Vote Share in Swing States Over Time",
    x = "Election Year",
    y = "Democratic Vote Share",
    color = "State"
  ) +
  theme_minimal() +
  scale_y_continuous(labels = scales::percent_format())

```

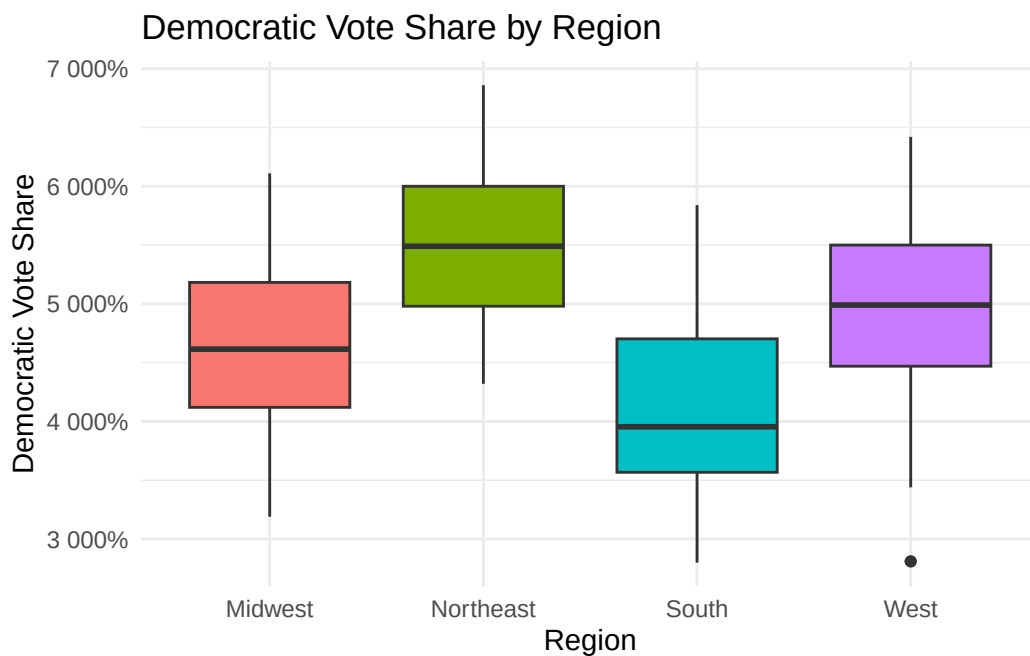


Example prompts:

- “Create a line chart showing Democratic vote share trends over time for swing states”
- “Show how turnout has changed from 2000-2020 by region”
- “Plot vote margin trends for competitive states”

2.1.2 5.2 Geographic Patterns

```
# Example: Compare regions
ggplot(election_data) +
  aes(x = region, y = dem_vote_share, fill = region) +
  geom_boxplot() +
  labs(
    title = "Democratic Vote Share by Region",
    x = "Region",
    y = "Democratic Vote Share"
  ) +
  theme_minimal() +
  theme(legend.position = "none") +
  scale_y_continuous(labels = scales::percent_format())
```



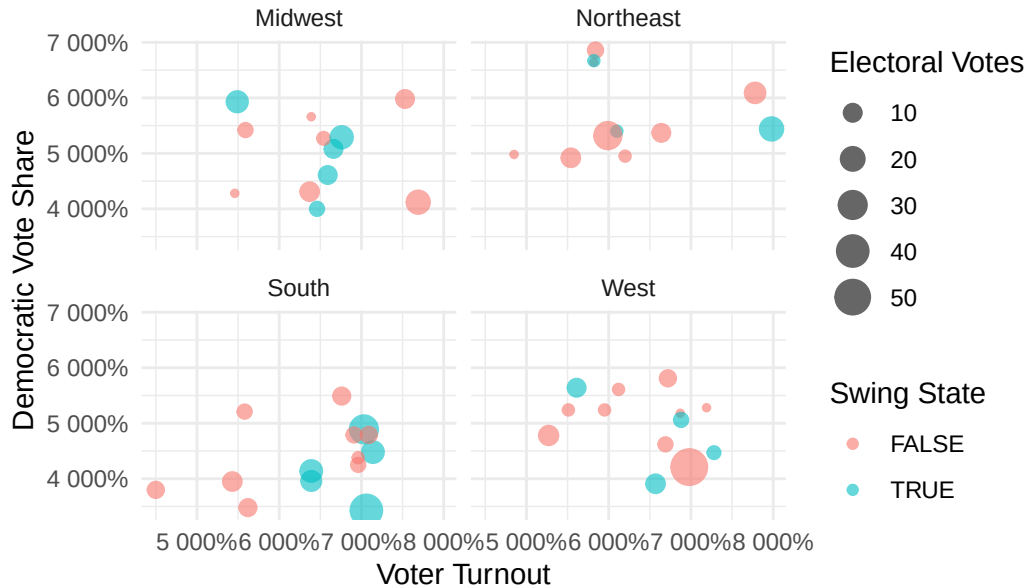
Example prompts:

- “Create boxplots comparing Democratic vote share across regions”
- “Make a scatter plot with electoral votes on the x-axis and vote margin on y-axis”
- “Show swing states vs non-swing states with different colors”

2.1.3 5.3 Multi-dimensional Display

```
# Example: Multiple variables simultaneously
ggplot(election_data %>% filter(year == 2020)) +
  aes(x = turnout, y = dem_vote_share,
      size = electoral_votes, color = swing_state) +
  geom_point(alpha = 0.6) +
  facet_wrap(~region) +
  labs(
    title = "2020 Election: Turnout, Vote Share, and Electoral Votes by Region",
    x = "Voter Turnout",
    y = "Democratic Vote Share",
    size = "Electoral Votes",
    color = "Swing State"
  ) +
  theme_minimal() +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format())
```


2020 Election: Turnout, Vote Share, and Electoral Votes by I



Example prompts:

- “Create a scatter plot showing turnout vs vote share, sized by electoral votes, faceted by region”
- “Show multiple dimensions using color, size, and facets”

2.2 6. Survey Response Patterns

Dataset: data/survey__visualization.csv

Description: Large political survey with demographic breakdowns.

Variables: - **respondent_id:** Unique identifier (int) - **age_group:** Age category (chr) - **education:** Education level (chr) - **income_bracket:** Income category (chr) - **party_id:** Party identification (chr) - **ideology:** Liberal-conservative scale 1-7 (int) - **approval_president:** Presidential approval 0-100 (dbl) - **approval_congress:** Congressional approval 0-100 (dbl) - **trust_media:** Media trust 0-100 (dbl) - **political_interest:** Interest level 1-10 (int)

2.2.1 6.1 Distribution Visualizations

```
# Load the dataset
survey_data <- read_csv("data/survey_visualization.csv")
```

```

Rows: 1000 Columns: 10
-- Column specification -----
Delimiter: ","
chr (4): age_group, education, income_bracket, party_id
dbl (6): respondent_id, ideology, approval_president, approval_congress, tru...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```

# Check the structure
glimpse(survey_data)

```

```

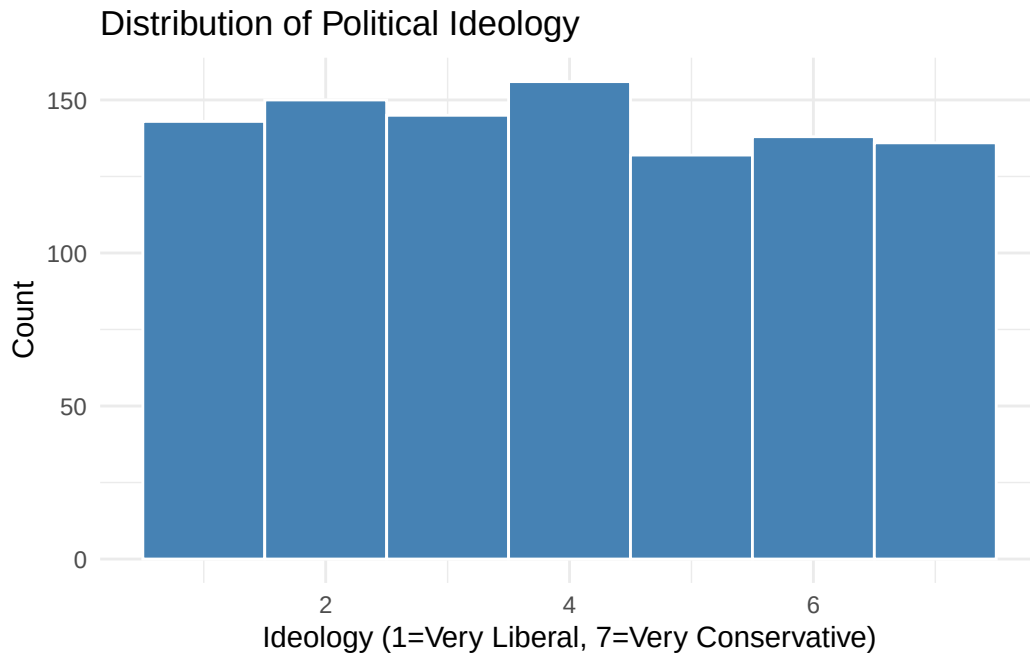
Rows: 1,000
Columns: 10
$ respondent_id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ~
$ age_group          <chr> "65+", "65+", "65+", "45-64", "65+", "18-29", "45-6~
$ education           <chr> "Some College", "HS", "Graduate", "HS", "Graduate",~
$ income_bracket      <chr> "30-60k", "30-60k", "<30k", "<30k", ">100k", "30-60~
$ party_id           <chr> "Republican", "Independent", "Republican", "Democra~
$ ideology            <dbl> 3, 3, 4, 3, 2, 7, 7, 3, 6, 7, 1, 1, 7, 5, 6, 7, 6, ~
$ approval_president <dbl> 9.5, 16.7, 34.0, 0.0, 36.4, 0.0, 33.6, 61.6, 37.4, ~
$ approval_congress  <dbl> 10.0, 14.1, 16.1, 56.1, 25.1, 4.2, 52.2, 10.2, 54.9~
$ trust_media         <dbl> 58.0, 45.2, 21.7, 51.1, 46.4, 38.8, 49.5, 37.2, 43.~
$ political_interest <dbl> 10, 9, 3, 3, 1, 1, 8, 3, 8, 2, 2, 6, 6, 3, 2, 4, 2,~

```

```

# Example: Ideology distribution
ggplot(survey_data) +
  aes(x = ideology) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of Political Ideology",
    x = "Ideology (1=Very Liberal, 7=Very Conservative)",
    y = "Count"
  ) +
  theme_minimal()

```

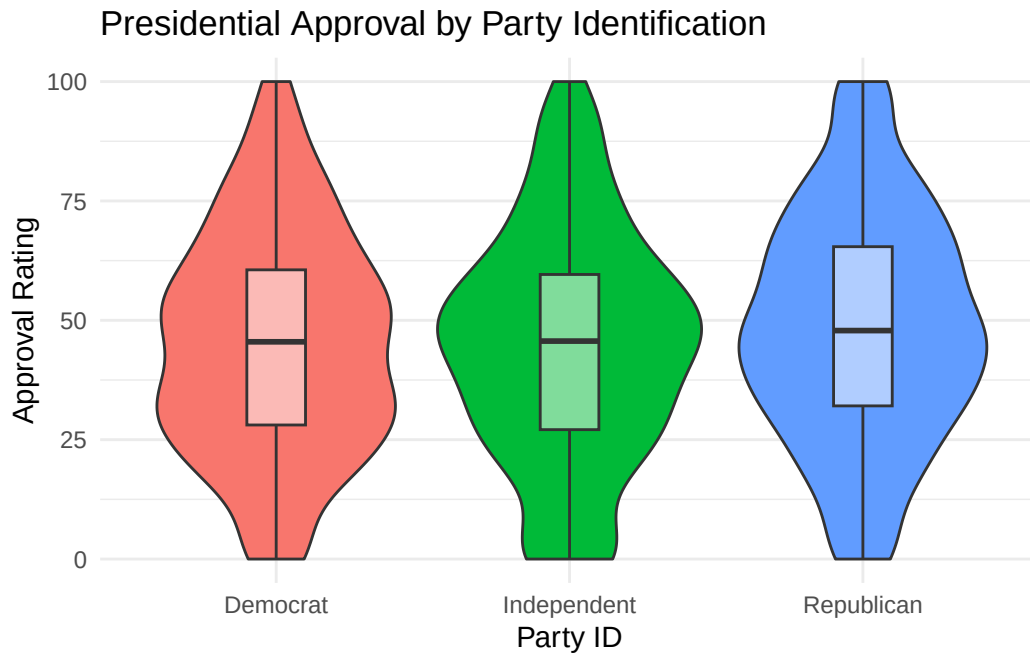


Example prompts:

- “Create a histogram of presidential approval ratings”
- “Show the distribution of political interest with a density plot”
- “Make a histogram of ideology scores”

2.2.2 6.2 Group Comparisons

```
# Example: Approval by party
ggplot(survey_data) +
  aes(x = party_id, y = approval_president, fill = party_id) +
  geom_violin() +
  geom_boxplot(width = 0.2, fill = "white", alpha = 0.5) +
  labs(
    title = "Presidential Approval by Party Identification",
    x = "Party ID",
    y = "Approval Rating"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



Example prompts:

- “Create violin plots comparing presidential approval across party identification”
- “Make boxplots showing trust in media by education level”
- “Compare ideology distributions across age groups using ridgeline plots”

2.2.3 6.3 Correlation Display

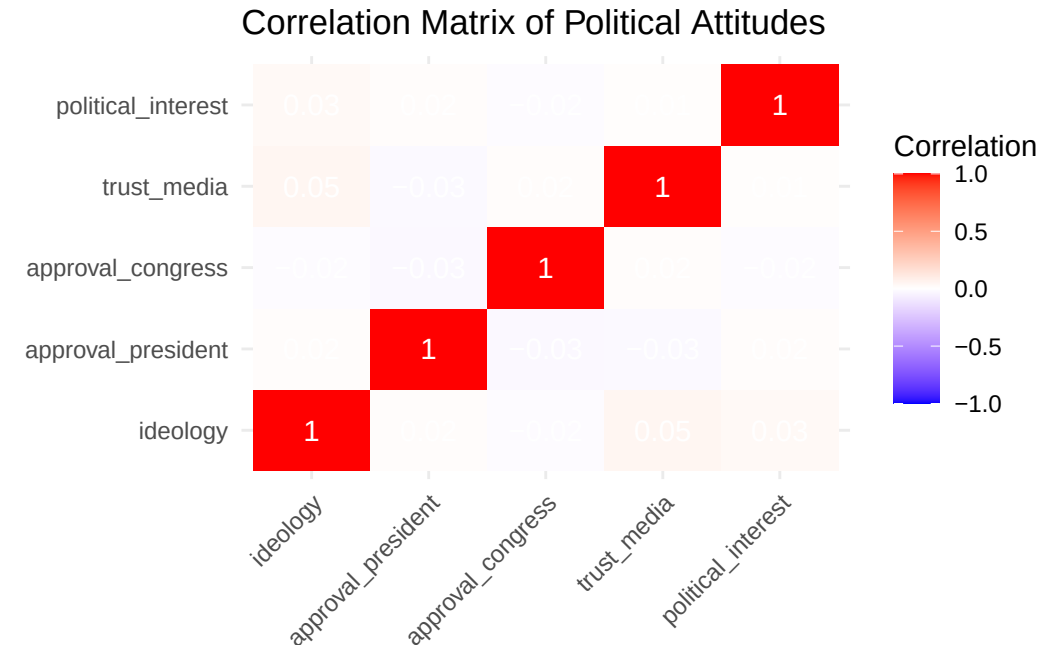
```
# Example: Correlation heatmap
cor_data <- survey_data %>%
  select(ideology, approval_president, approval_congress,
         trust_media, political_interest) %>%
  cor() %>%
  melt()

ggplot(cor_data) +
  aes(x = Var1, y = Var2, fill = value) +
  geom_tile() +
  geom_text(aes(label = round(value, 2)), color = "white") +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red",
                      midpoint = 0, limits = c(-1, 1)) +
  labs(
```

```

    title = "Correlation Matrix of Political Attitudes",
    x = NULL, y = NULL,
    fill = "Correlation"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



Example prompts:

- “Create a correlation heatmap of all numeric variables”
- “Make a scatterplot matrix showing relationships between approval ratings and ideology”
- “Show the correlation between media trust and presidential approval”

2.3 7. Legislative Behavior

Dataset: data/legislative_visualization.csv

Description: Congressional voting patterns and member characteristics.

Variables: - **member_id**: Legislator identifier (int) - **name**: Member name (chr) - **party**: Political party (chr) - **state**: State abbreviation (chr) - **chamber**: House or Senate (chr) - **ideology_score**: DW-NOMINATE score (dbl) - **votes_with_party**: Percentage voting with party (dbl) - **bills_sponsored**: Number of bills sponsored (int) - **committee_leadership**: Leadership positions (int) - **tenure**: Years in office (int)

2.3.1 7.1 Ideology Visualization

```
# Load the dataset
legislative_data <- read_csv("data/legislative_visualization.csv")
```

Rows: 535 Columns: 10

-- Column specification -----

Delimiter: ","

chr (4): name, party, state, chamber

dbl (6): member_id, ideology_score, votes_with_party, bills_sponsored, commi...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Explore the data
glimpse(legislative_data)
```

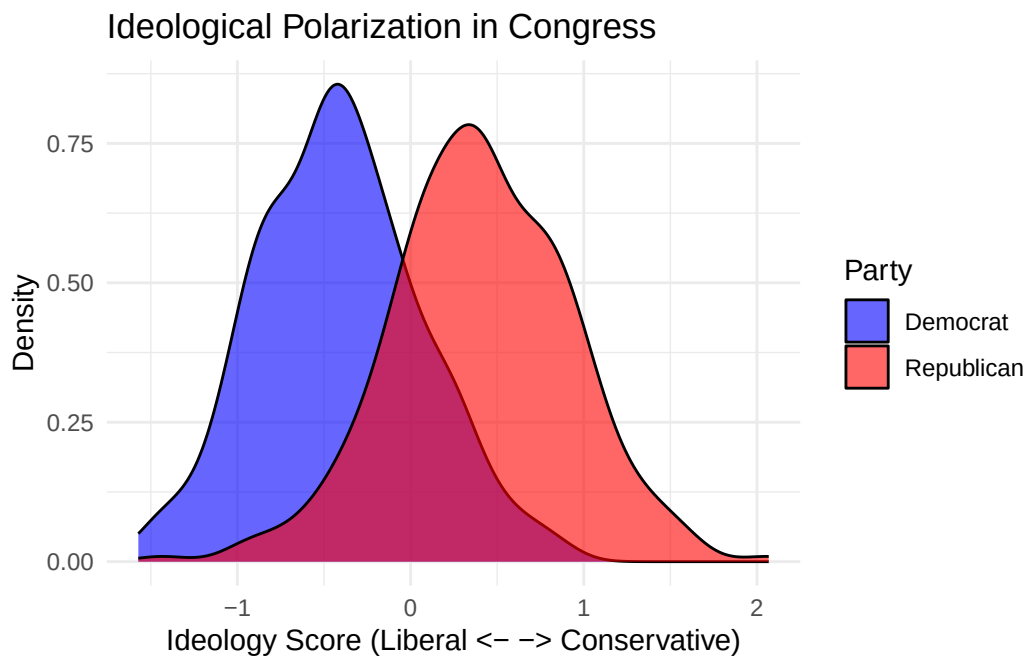
Rows: 535

Columns: 10

\$ member_id	<dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15~
\$ name	<chr> "Member 1", "Member 2", "Member 3", "Member 4", "~
\$ party	<chr> "Democrat", "Democrat", "Democrat", "Democrat", "~
\$ state	<chr> "IL", "VT", "OH", "KY", "IN", "GA", "AZ", "AZ", "~
\$ chamber	<chr> "House", "House", "House", "House", "House", "Hou~
\$ ideology_score	<dbl> -0.024, -0.287, -1.451, -0.294, 0.434, 0.668, 0.1~
\$ votes_with_party	<dbl> 92.2, 90.7, 89.6, 78.6, 84.3, 84.3, 95.9, 98.8, 8~
\$ bills_sponsored	<dbl> 14, 49, 22, 18, 41, 14, 23, 30, 48, 12, 7, 23, 25~
\$ committee_leadership	<dbl> 0, 0, 0, 0, 0, 0, 1, 0, 2, 0, 1, 0, 1, 0, 0, 0, 0~
\$ tenure	<dbl> 22, 30, 6, 39, 38, 33, 34, 29, 6, 29, 24, 25, 19, ~

```
# Example: Ideology distribution by party
ggplot(legislative_data) +
  aes(x = ideology_score, fill = party) +
  geom_density(alpha = 0.6) +
  labs(
    title = "Ideological Polarization in Congress",
    x = "Ideology Score (Liberal ← → Conservative)",
    y = "Density",
    fill = "Party"
  ) +
```

```
theme_minimal() +
scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red"))
```



Example prompts:

- “Create density plots showing ideology distribution by party”
- “Make histograms comparing House and Senate ideology”
- “Show polarization using overlapping distributions”

2.3.2 7.2 Bivariate Relationships

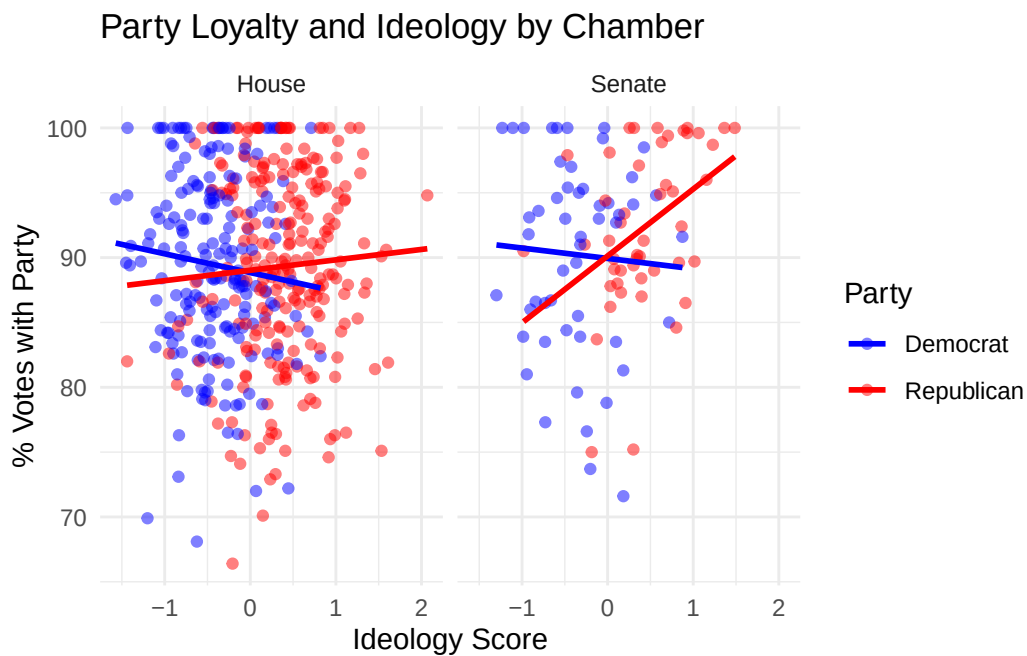
```
# Example: Party unity vs ideology
ggplot(legislative_data) +
  aes(x = ideology_score, y = votes_with_party, color = party) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~chamber) +
  labs(
    title = "Party Loyalty and Ideology by Chamber",
    x = "Ideology Score",
    y = "% Votes with Party",
```

```

    color = "Party"
  ) +
  theme_minimal() +
  scale_color_manual(values = c("Democrat" = "blue", "Republican" = "red"))

```

`geom\_smooth()` using formula = 'y ~ x'



Example prompts:

- “Create a scatterplot of ideology vs party unity, colored by party”
- “Show the relationship between tenure and bills sponsored”
- “Compare chambers using facets”

2.3.3 7.3 Complex Comparisons

```

# Example: Multi-dimensional comparison
ggplot(legislative_data) +
  aes(x = tenure, y = bills_sponsored,
      color = ideology_score, size = committee_leadership + 1) +
  geom_point(alpha = 0.6) +

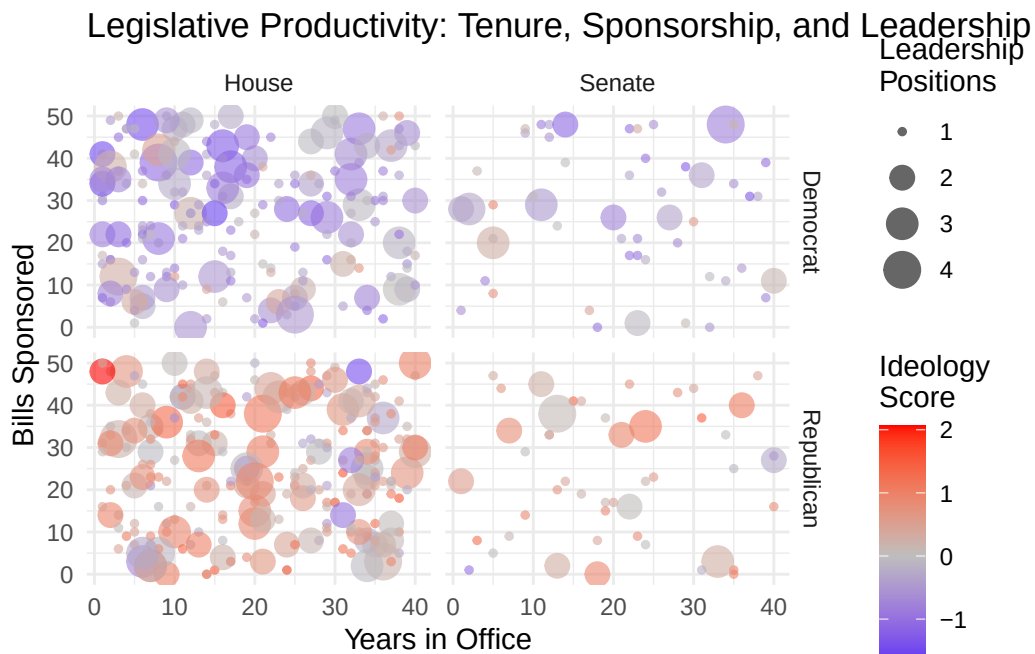
```



```

facet_grid(party ~ chamber) +
labs(
  title = "Legislative Productivity: Tenure, Sponsorship, and Leadership",
  x = "Years in Office",
  y = "Bills Sponsored",
  color = "Ideology\nScore",
  size = "Leadership\nPositions"
) +
theme_minimal() +
scale_color_gradient2(low = "blue", mid = "gray", high = "red", midpoint = 0)

```



Example prompts:

- “Create a comprehensive plot showing tenure, productivity, ideology, and leadership”
- “Make a publication-quality visualization comparing all aspects of legislative behavior”